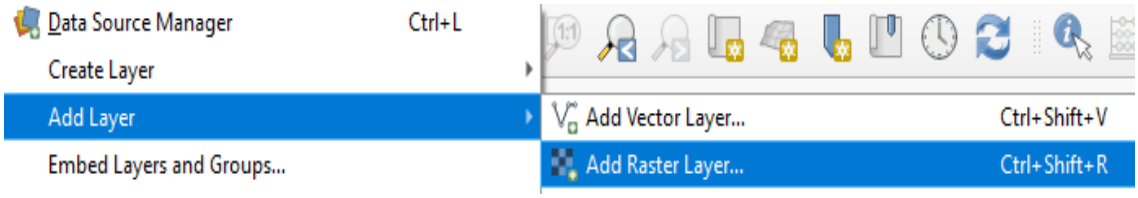
Practical 3

Exploring and Managing Raster data: Adding raster layers, raster styling and analysis, raster mosaicking and clipping

❖ Exploring and Managing Raster Data:

A. Adding raster layers

➢ From menu bar Select Layer→Add Layer→Add Raster Layer



➢ Select Gridded Population of the World (GPW) v3 dataset from Columbia University, Population Density Grid for the entire globe in ASCII format and for the year 1990 and 2000.
"\GIS_Workshop\Practicals\Practical_02\A\Data\gl_gpwv3_pdens_90_ascii_one\glds90ag60.asc"
"\GIS_Workshop\Practicals\Practical_02\A\Data\gl_gpwv3_pdens_90_ascii_one\glds00ag60.asc"

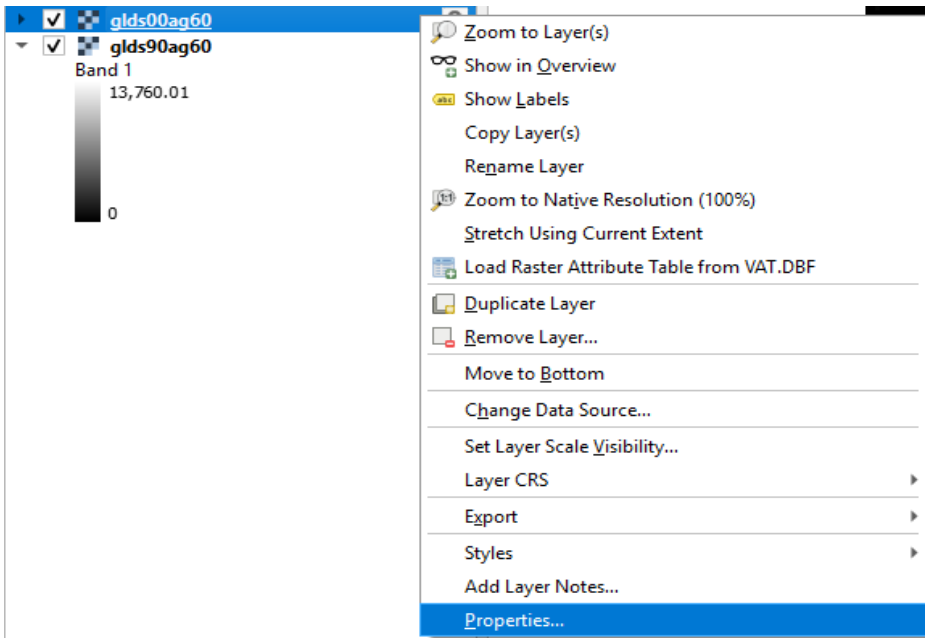
➢ Go to Project→Properties or Press the

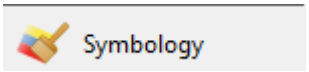
Set CRS option on bottom right corner.
Select WGS 84 EPSG: 4326 and Press OK.

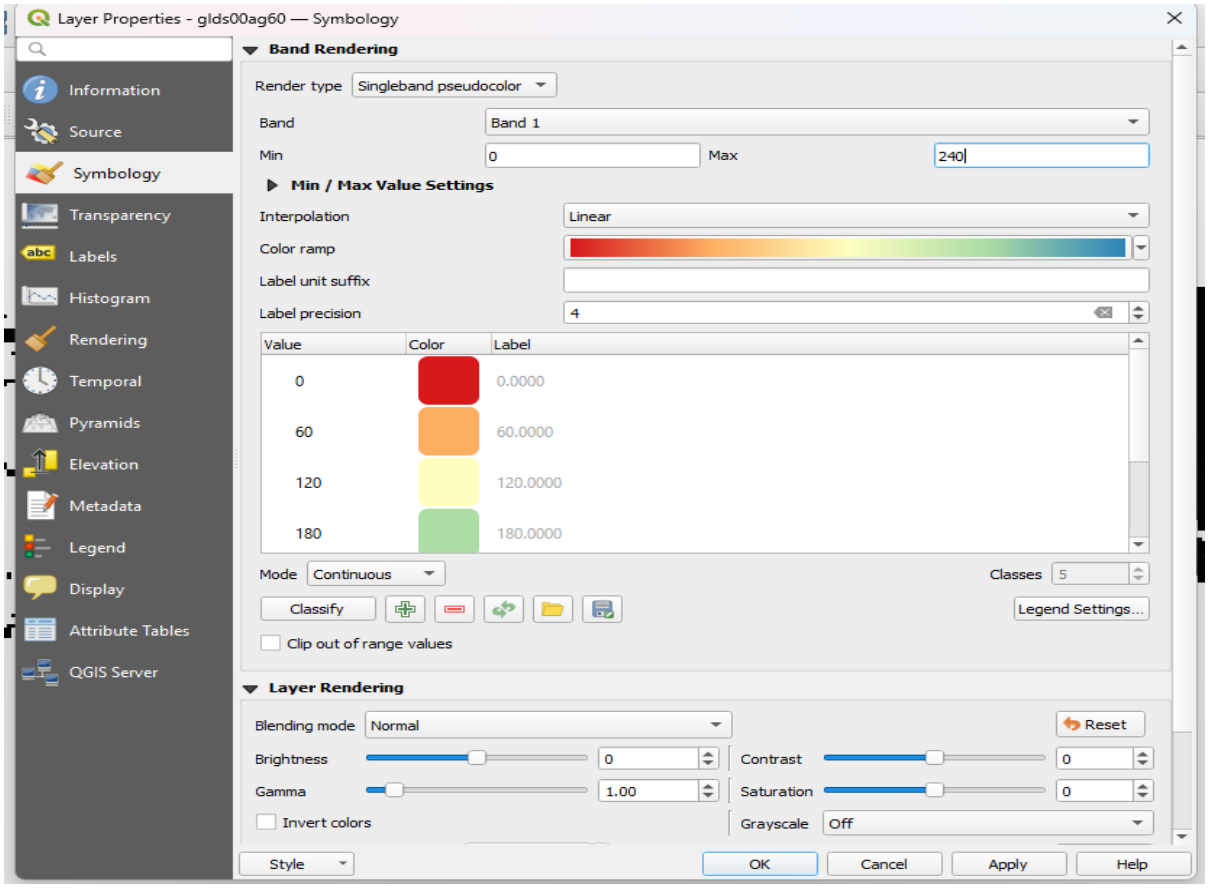
B. Raster Styling and Analysis

➢ To start with analysis of population data, convert the pixel from grayscale to Color.

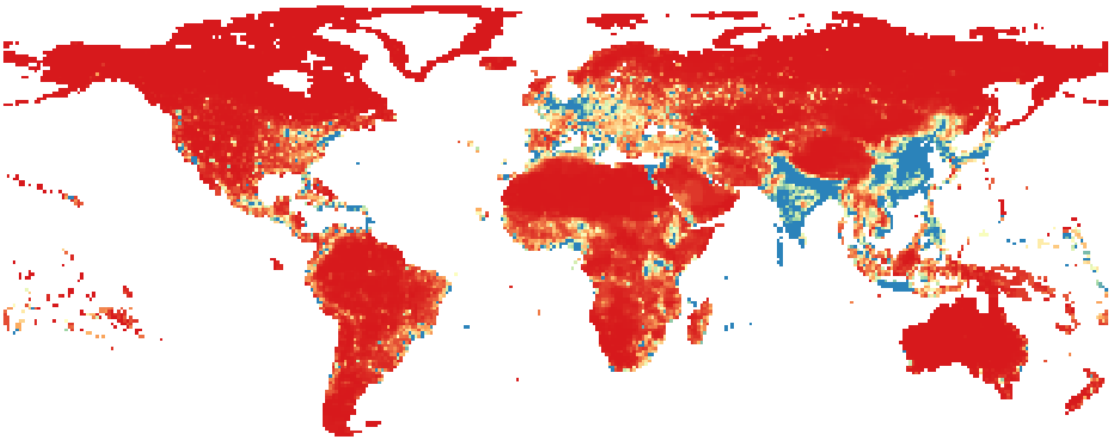
➢ Select "glds90ag60.asc" Layer from layer Pane→Select Property OR Double Click on it.



- Select 

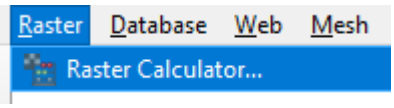


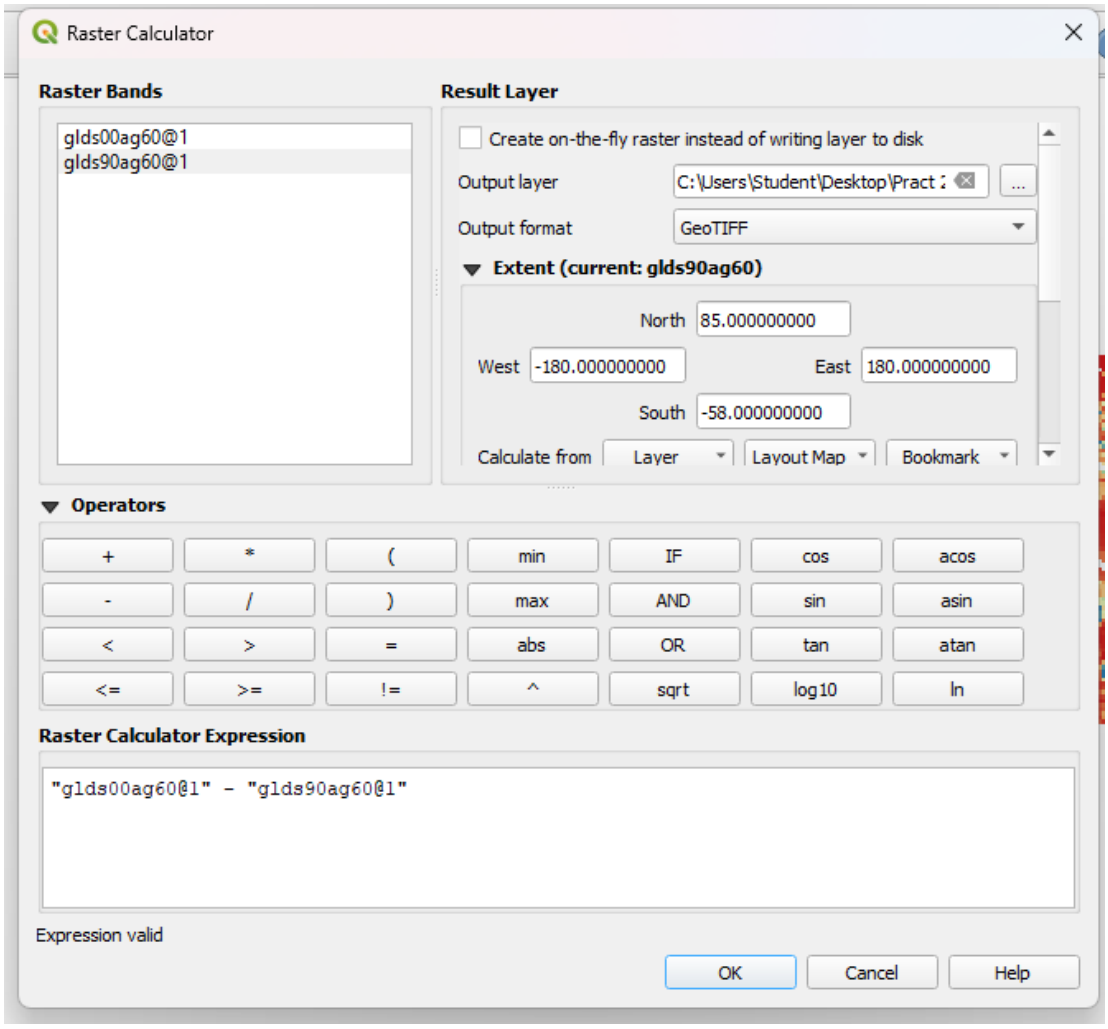
- Press “APPLY”
- Repeat the same for “glds00asc.60” Layer

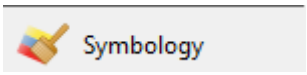


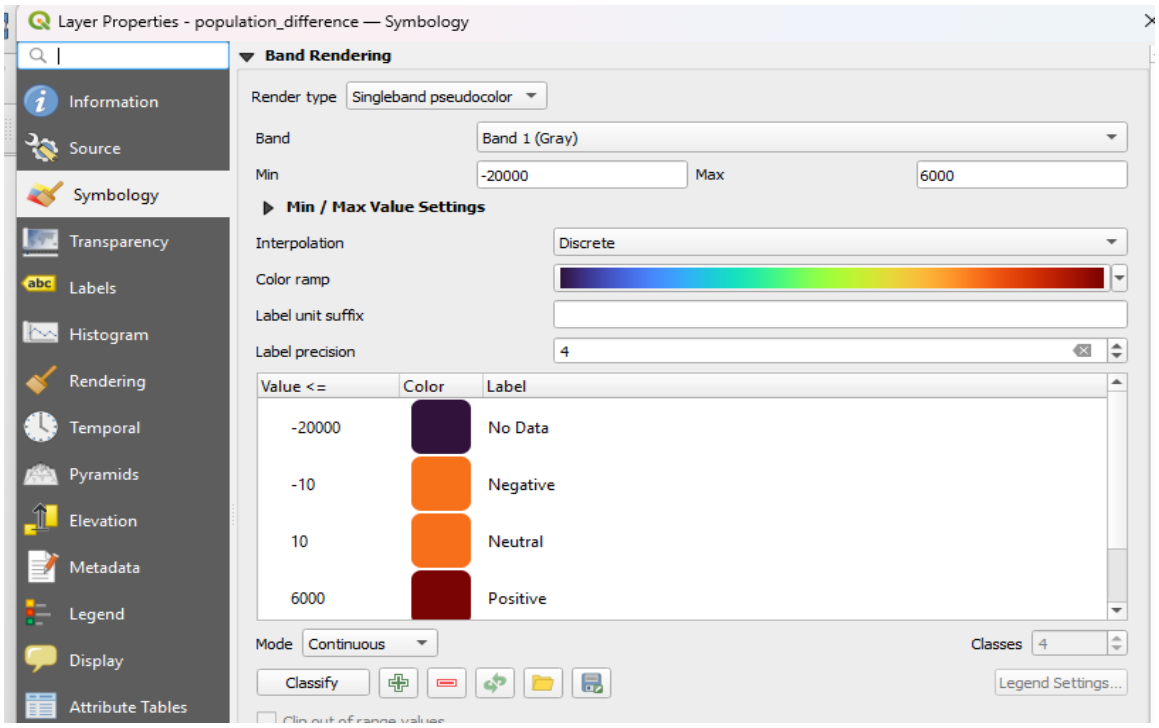
Layer output after applying style.

- The objective this experiment is to analyse raster data, as an example we will find areas with largest population change between 1990 and 2000, by calculating the difference between each pixel values.
- Go to Raster→Raster Calculator

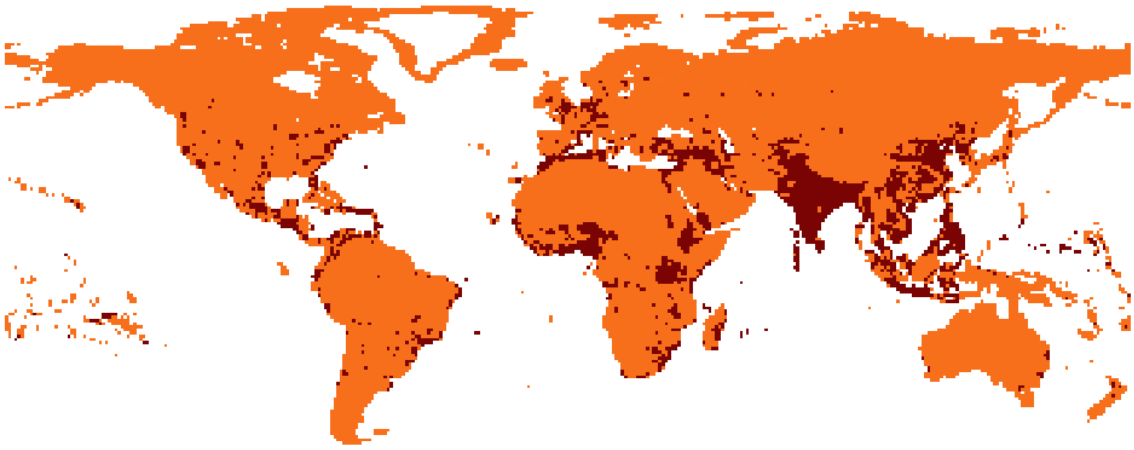




- Put the expression “glds00ag60@1”- “glds90ag60@1”
- Select the output file location & name and press OK.
- Remove the other two layers i.e. glds00ag60.asc and glds90ag60.asc
- Double click on pop_diff layer.
- Select 



- Set render type to “single band pseudo color” Interpolation as discrete, and remove all classification and add as shown in figure above using button. After all settings press “OK”.
- Layer will appear like



Value <=	Color	Label
-20000		No Data
-10		Negative
10		Neutral
6000		Positive

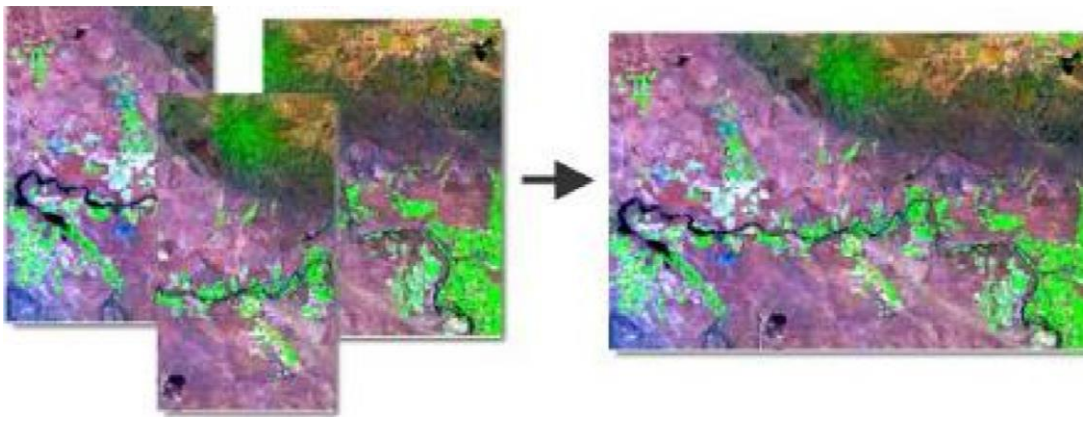
- Explore an area of your choice and check the raster band value using to verify the classification rule.
- The Orange pixel shows negative changes and purple show positive changes.

C. Raster Mosaicking and Clipping

A mosaic is a combination or merge of two or more images.

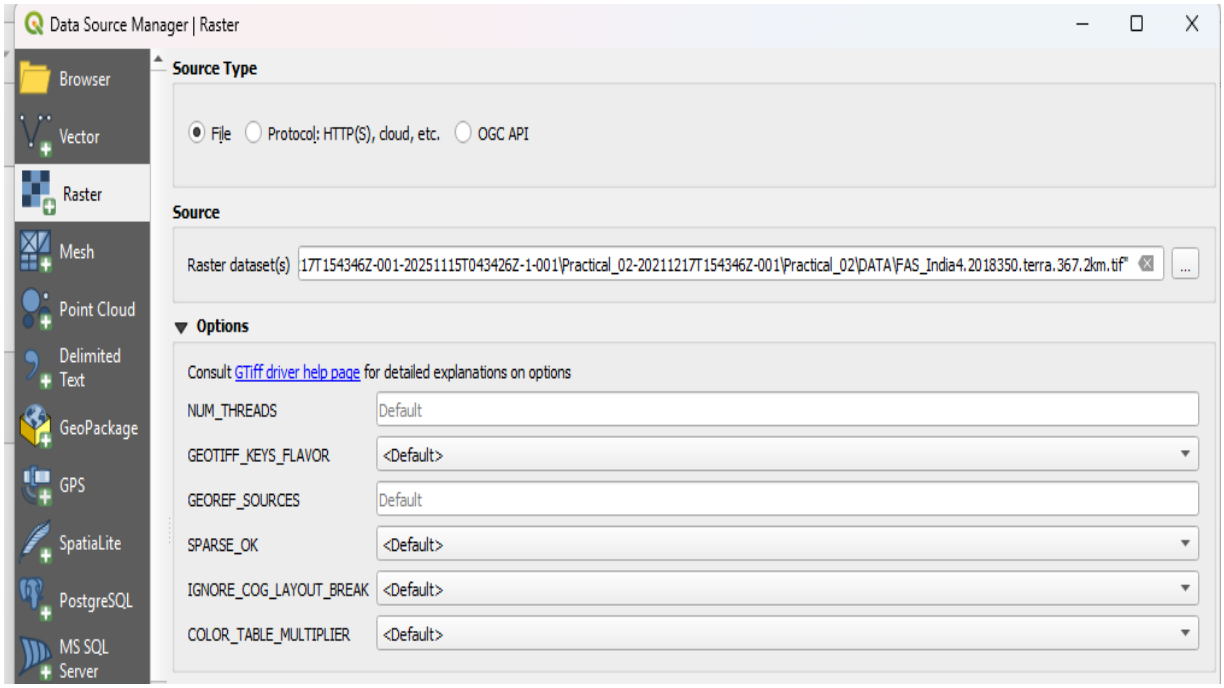
In GIS, a single raster dataset can be created from multiple raster datasets by mosaicking them together.

In many cases, there will be some overlap of the raster dataset edges that are being mosaicked together, as shown below.

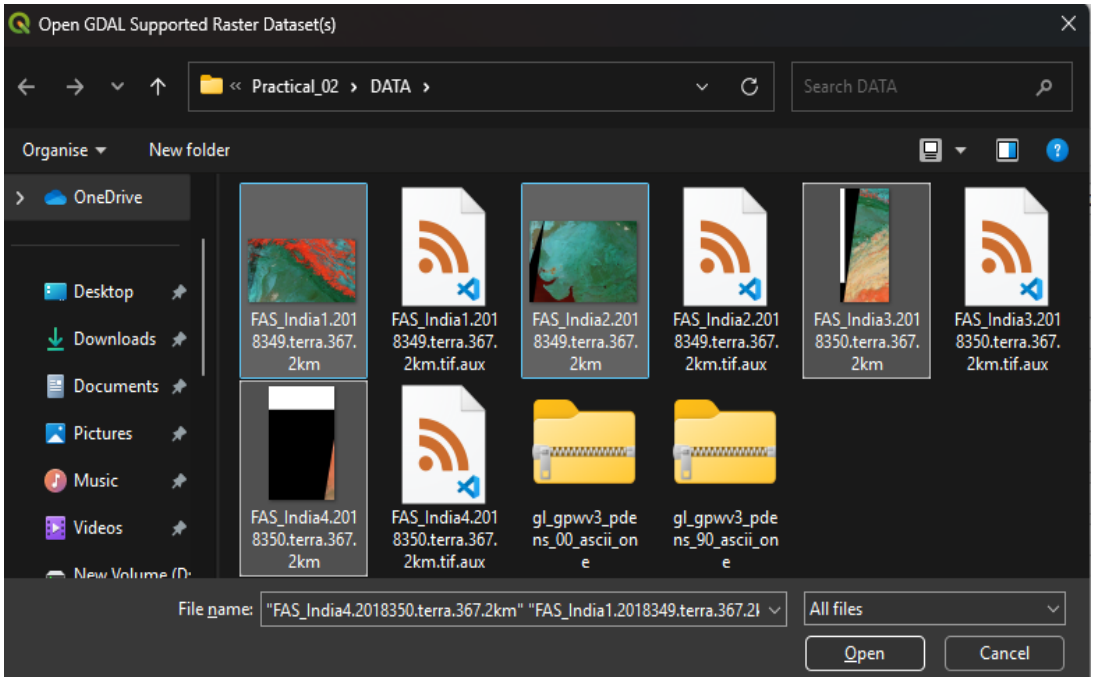


These overlapping areas can be handled in several ways: for example, you can choose to only keep raster data from the first or last dataset, you can blend the overlapping cell values using a weight based algorithm, you can take mean of the overlapping cell values, or you can take the minimum or maximum value. When mosaicking discrete data, the First, Minimum, or Maximum options give the most meaningful results. The Blend and Mean options are best suited for continuous data. If any of the input raster are floating point, the output is floating point. If all the inputs are integer and First, Minimum, or Maximum is used, the output is integer.

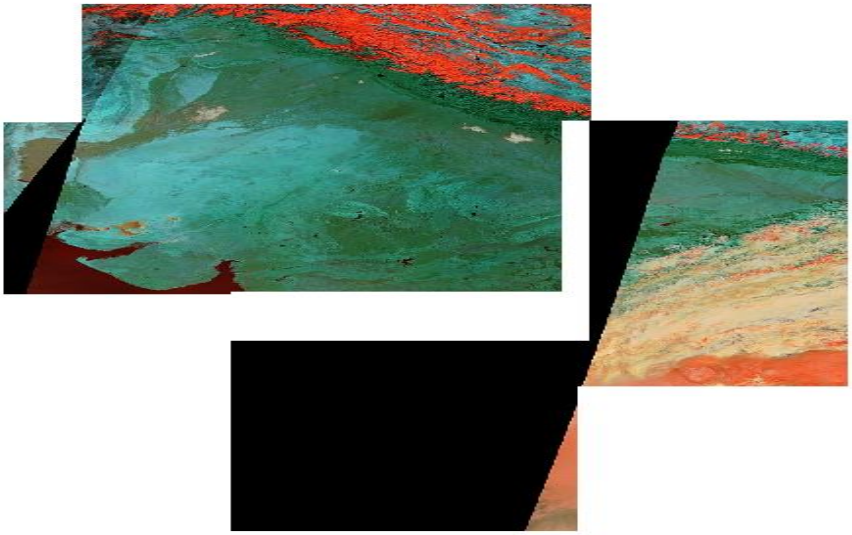
- Go to Layer→Add Layer→Add Raster Layer.



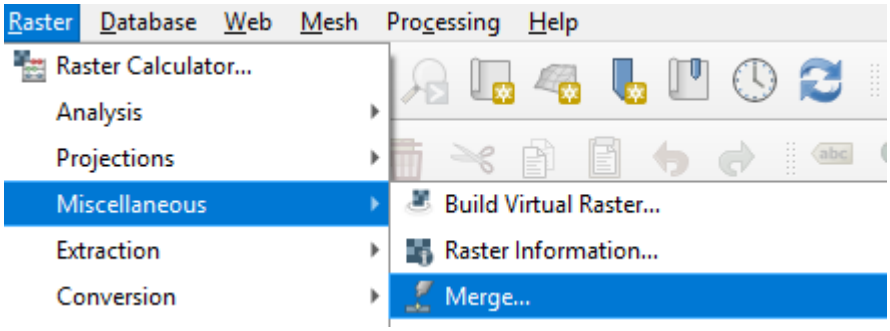
- Select the following "**.tif** " raster image for India from data folder.
  FAS_India1.2018349.terra.367.2km.tif
  FAS_India2.2018349.terra.367.2km.tif
  FAS_India3.2018349.terra.367.2km.tif
  FAS_India4.2018349.terra.367.2km.tif



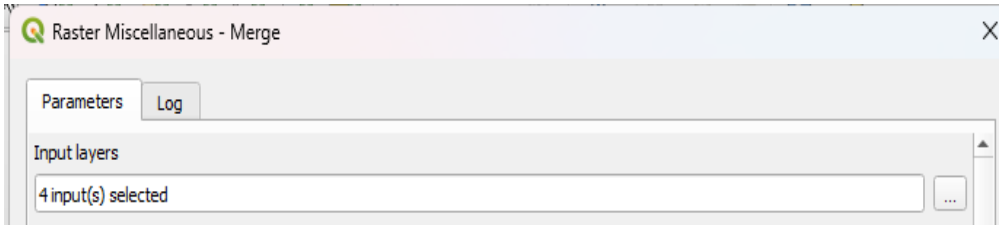
- Press Open.
- In data source manager | Raster window click Add.



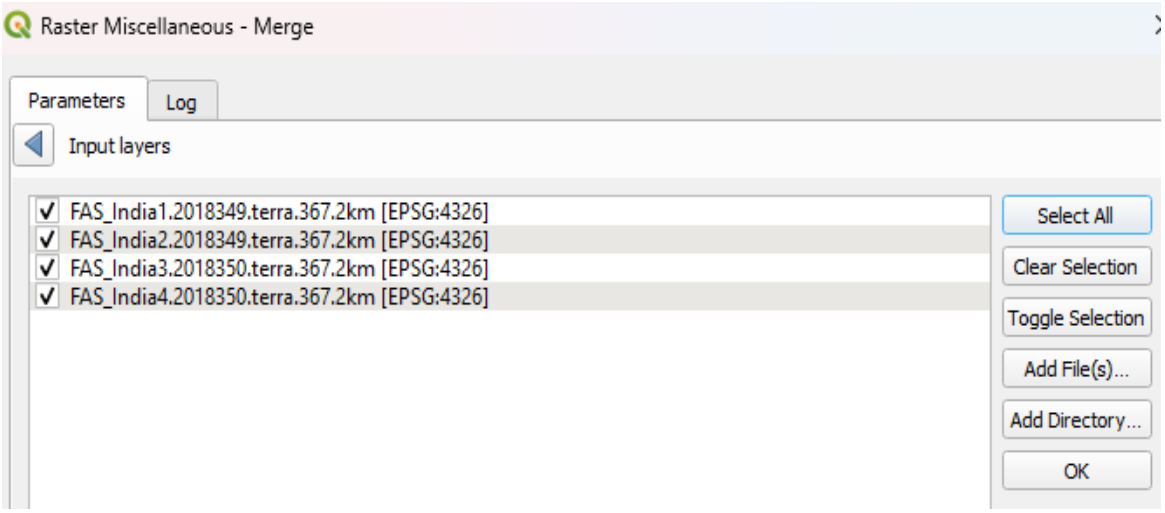
- Go to Raster→Miscellaneous→Merge



- In the Merge dialog window

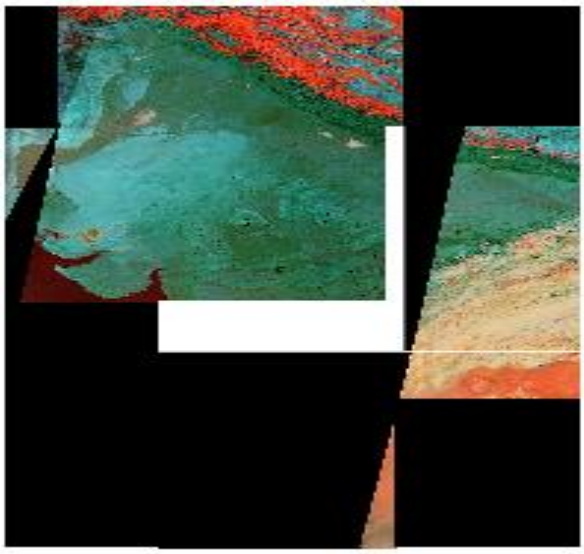


- Select all layers and Press OK.

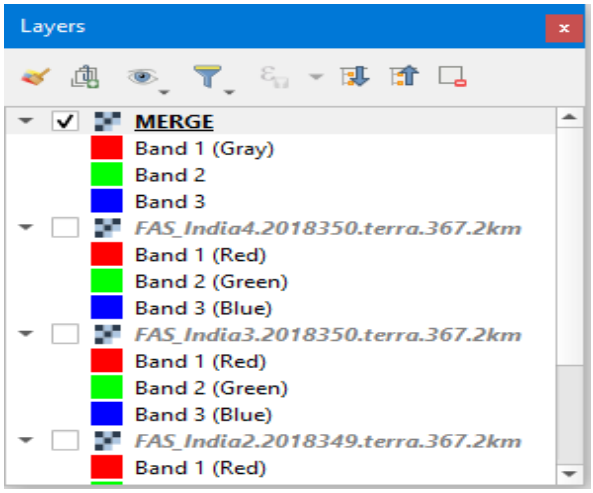


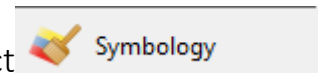
- In Merge dialog window select a file name and location to save merge images.

- Save the file to "GIS_Workshop /Practical's/Practical_02/C/" location with the name as Merge_Files.tif
- Press Run and after completion of operation close the Merge window dialog box.



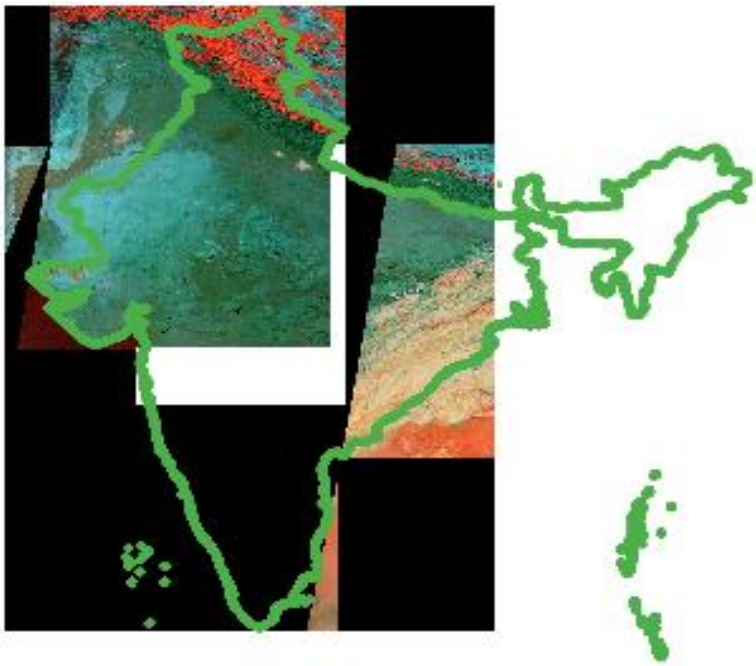
- You can now deselect individual layers from layer pane and only keep the merged raster file.



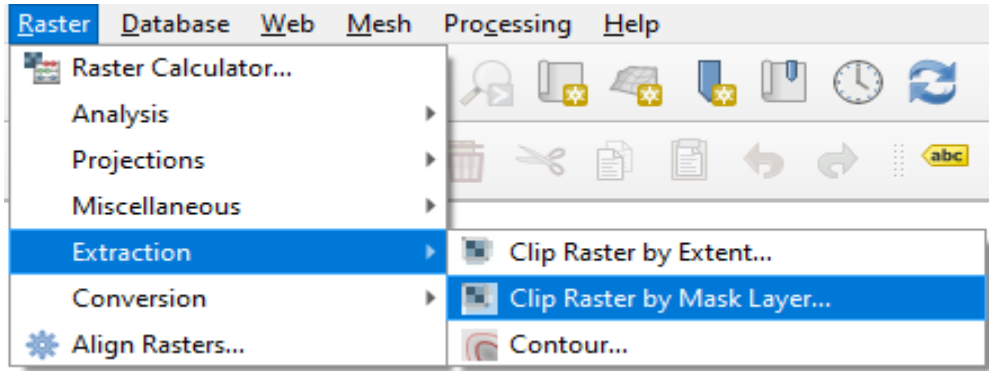
- Go to Layer→Add Vector Layer→Select \GIS_Workshop\Practicals\Practical_02\IndiaAdminBoundry\IND_adm0.shp file.
- From layer properties→Select 

→Select any one of the following

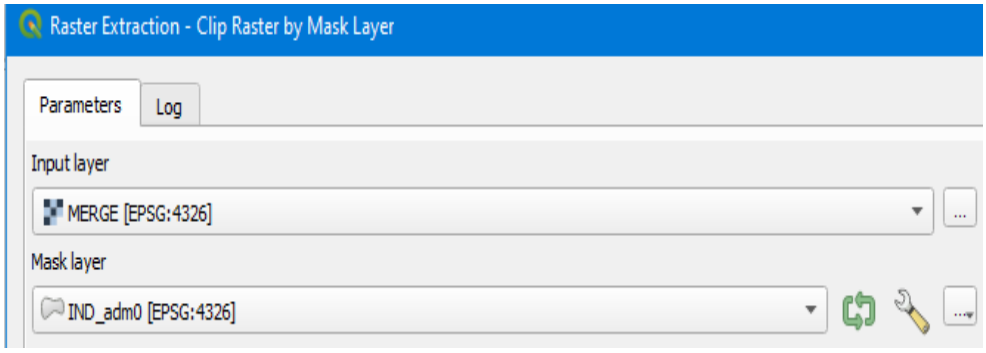
- The result will be



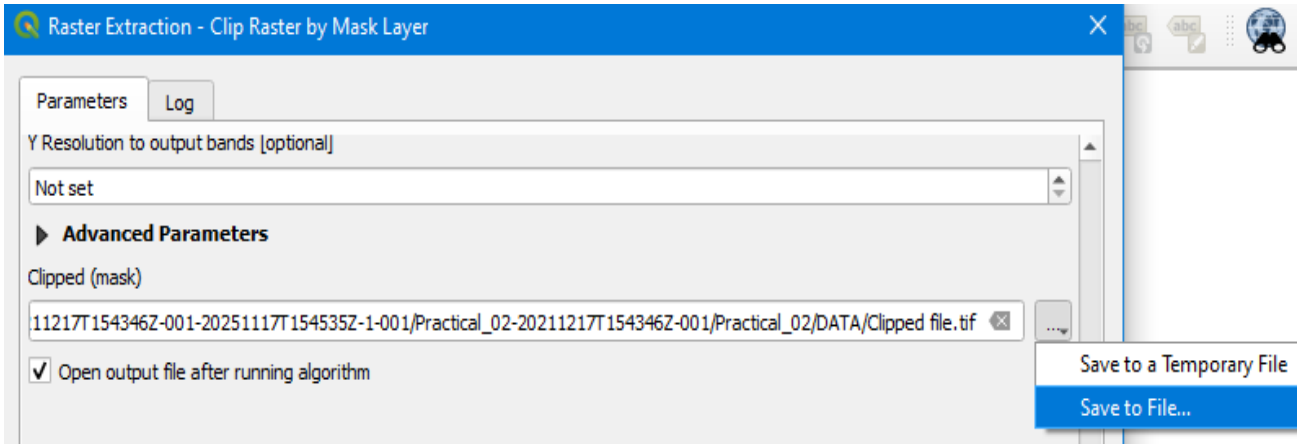
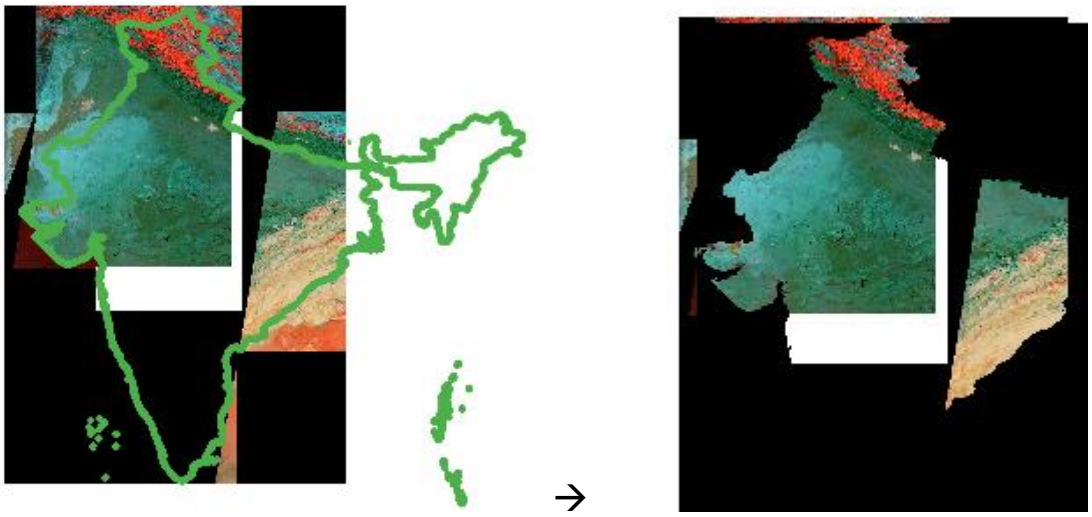
- Go to Raster→Extraction→Clip Raster by Mask Layer



- Select the merge raster image as input and Ind_adm0 as mask layer.



- Select a file name and location for clipped raster as/Practical_02/C/Clipped_File.tif.



- Press RUN.